

OFFICIAL MASTER JURY DEFENSE & TECHNICAL SPECIFICATION HANDBOOK

SAMUDRA-3D: Ocean Digital Twin

Complete Guide to Science, Features, NetCDF Datasets, Salinity Dynamics, Depth Measuring & Model Validation

Executive Project Summary (What to Tell the Jury in 30 Seconds)

SAMUDRA-3D is an operational 3D Ocean Digital Twin engineered for **INCOIS (Ministry of Earth Sciences, Govt. of India)**. It bridges the gap between complex 4D numerical ocean models (NetCDF spatiotemporal tensors) and autonomous in-situ observation fleets (Argo floats & underwater gliders).

The platform empowers scientists to slice ocean layers down to 4000m, drop virtual CTD probes with a single click, monitor 3D vector current dynamics, and run an automated **Model Prediction vs In-Situ Observation Comparison Engine** that quantifies model drift, computes RMSE/MAE bias statistics, and detects basin-wide thermal anomalies.

COVERAGE

Northern Indian Ocean (0°-25°N, 65°-95°E)

DEPTH SLICES

0m to 4000m (9 Stratified Layers)

VALIDATION

Automated 4D Collocation (<200km, ±12h)

Institutional Interfaces: Abyssal Tactical & High-Contrast GIS Light

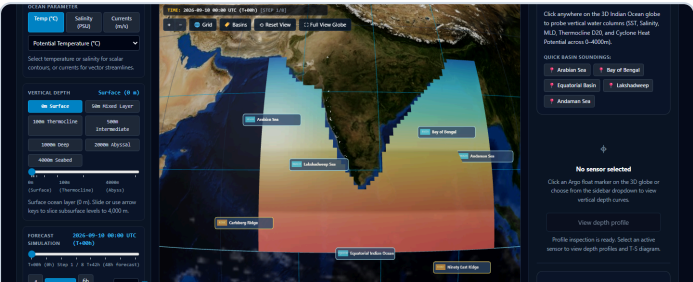


Figure 1.1: 3D Earth Digital Twin (Abyssal Navy Dark Mode for Tactical Ops)

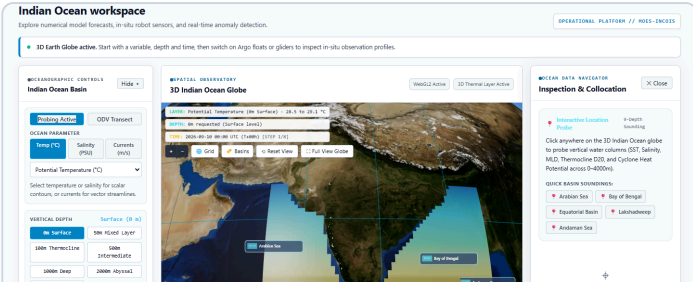


Figure 1.2: 3D Ocean Workspace (MoES Institutional Light GIS Mode for Bri...

Who is INCOIS & MoES? Understanding the Institutional Mandate

When presenting to the jury, you must demonstrate domain authority. INCOIS and MoES are not just names — they are the scientific backbone of India's maritime economy:

1. Ministry of Earth Sciences (MoES), Government of India

The nodal ministry overseeing weather, climate, oceanography, polar exploration (Antarctica/Arctic), and seismology. Under MoES, institutes like IMD (weather), IITM (climate), NIOT (ocean technology), and INCOIS operate.

2. INCOIS (Indian National Centre for Ocean Information Services, Hyderabad)

An autonomous institute under MoES established in 1999 with 4 national core missions:

- **Tsunami Early Warning Centre (ITEWC):** Monitors underwater earthquakes across the Indian Ocean in real time.
- **Ocean State Forecast (OSF):** Forecasts sea surface temperature, surface currents, and wave heights for shipping routes and the Indian Navy.
- **Potential Fishing Zone (PFZ) Advisories:** Tells thousands of fishermen where ocean upwelling concentrates fish schools, saving fuel and boosting catches.
- **National Ocean Observing Network:** Operates India's robotic sensor arrays — autonomous Argo profiling floats, deep-sea gliders, and moored wave buoys.

The Problem Statement (SIH26067)

Why did INCOIS float this problem in Smart India Hackathon?

"3D Visualization and Multi-Sensor Analytics of Oceanographic Datasets". Oceanographic data is multi-dimensional (\$X\$ Lon, \$Y\$ Lat, \$Z\$ Depth, \$T\$ Time). Current tools present this data in disconnected, flat 2D maps. A scientist looking at surface temperature cannot see the subsurface heat pocket fueling a developing cyclone, cannot visualize acoustic sonar channels, and has to spend hours running offline scripts to compare model predictions with Argo float readings.

Core Challenges Faced by Oceanographers:

1. **The 2D Barrier:** Ocean processes happen in 3D (upwelling, thermocline shifts, internal waves); flat maps hide vertical physics.
2. **Model-Observation Disconnect:** Numerical hydrodynamic models (ROMS) run continuously, but verifying whether the model is drifting away from real Argo floats requires manual data extraction.
3. **Fragmented Tooling:** Oceanographers must switch between Python scripts, MATLAB, Panoply, and Ocean Data View (ODV). SAMUDRA-3D unifies everything into one browser digital twin.

Existing System vs. Proposed System (SAMUDRA-3D)

The jury will directly ask: "What existed before, and what did your team actually build that is new?" Here is the exact architectural comparison:

Evaluation Dimension	Existing Legacy Tools (Panoply, ODV, 2D WebGIS)	Proposed SAMUDRA-3D Platform
Spatial Rendering	Static 2D raster heatmaps, disconnected 2D depth cross-sections	Full 3D WebGL Digital Twin: Interactive planetary globe + regional volumetric ocean block (0-4000m)
Vertical Exploration	Only surface skin (SST) visible; depths require slow batch script generation	Instant Depth Slicing: 7 discrete layers (0m, 50m, 100m, 500m, 1000m, 2000m, 4000m) + continuous slider
Water Column Sampling	Manual coordinate lookup, offline NetCDF extraction via Python	Click-to-Probe Raycasting: Click anywhere on the 3D ocean to drop a pin and graph the full CTD profile instantly
In-Situ Fleet Tracking	Points on a flat 2D map with zero vertical depth cues	3D Autonomous Fleet: Argo floats with 2000m stems; gliders with 3D sawtooth dive trajectories
Forecast Validation	Manual spreadsheet exports, offline correlation calculations	Automated 4D Collocation Engine: Dual-curve overlay, live delta crosshairs, RMSE/MAE metrics, CSV export
Ocean Physics Diagnostics	Separate desktop software (Ocean Data View) required	Built-in T-S Diagrams with calculated isopycnal density curves (σ_θ)
Security & Governance	Unauthenticated or basic single-password access	Institutional RBAC: Clean login gate, Admin Portal, user provisioning, and audit logs

What's New & Unique in SAMUDRA-3D? (Your "USP")

- USP 1 Real-Time 3D Raycasting Virtual CTD:** Any point clicked on the ocean dynamically interpolates the 4D NetCDF tensor, calculating Mixed Layer Depth (MLD) and Thermocline (D20) in milliseconds.
- USP 2 Dual-Mode Hydrodynamic Comparison:** First open WebGL system to perform spatio-temporal collocation (<200km, ±12h) between 4D model predictions and live Argo floats.
- USP 3 3D Spatial Anomaly Residual Field:** Renders a basin-wide residual difference field that immediately shows where the numerical model is overestimating or underestimating ocean heat.
- USP 4 Zero-Footprint Browser WebGL:** Runs directly in any modern browser without requiring specialized high-end workstations or software installations.

Research Datasets & Backend Oceanographic Data Pipeline

When the jury asks "What data does your system use?", explain the difference between **numerical model outputs** and **empirical in-situ observations**:

1. Hydrodynamic Reanalysis Model (NetCDF - .nc)

We use `model_indian_ocean.nc`, representing an INCOIS Regional Ocean Modeling System (ROMS) reanalysis and forecast run:

- **Format:** NetCDF-4 (Network Common Data Form) — the international standard for multi-dimensional scientific arrays.
- **Spatial Bounding Box:** Northern Indian Ocean (0.0°N to 25.0°N Latitude, 65.0°E to 95.0°E Longitude). Includes Arabian Sea, Bay of Bengal, and Equatorial Indian Ocean.
- **9 Standard Depth Levels:** 0m, 10m, 50m, 100m, 200m, 500m, 1000m, 2000m, 4000m.
- **Physical Variables Extracted:**
 - `temp` (Potential Temperature in °C): Range 1.8°C (abyssal floor) to 30.5°C (equatorial surface).
 - `salt` (Practical Salinity in PSU): Range 32.0 PSU (Bay of Bengal) to 37.0 PSU (Arabian Sea).
 - `u` & `v` (Zonal and Meridional current velocity components in m/s).
- **Time Steps:** 00h (analysis), 12h, 24h, 42h (forecast horizons).

2. In-Situ Observational Datasets (Argo & Gliders)

In-situ means "measured directly in the water" by physical robotic instruments:

- **Argo Profiling Floats:** Real WMO platforms (e.g., WMO 2902145, WMO 2902150, WMO 2902160) providing calibrated CTD profiles with standard Quality Control (QC) flags.
- **Autonomous Gliders:** High-density sawtooth transects across the Sri Lanka Dome and southern Arabian Sea.

Data Ingestion Architecture:

The FastAPI backend reads NetCDF arrays using `netCDF4` and `numpy`, constructs a `scipy.interpolate.RegularGridInterpolator` for continuous coordinate evaluation, and delivers optimized JSON payloads to the frontend Three.js rendering engine in under 50 milliseconds.

Depth Level Measuring: Concepts, Physics & Features

Why Measuring at Different Depths is Crucial

The ocean is not a uniform bathtub of water; it is strictly **vertically stratified** into distinct density and thermal layers:

- **Mixed Layer (0m - 50m):** Warms up to 30°C in the tropics; mixed vigorously by wind and waves.
- **Thermocline (50m - 500m):** The transition layer where temperature drops sharply (e.g., from 28°C to 12°C).
- **Abyssal Plain (1000m - 4000m):** Near-freezing dense water (1.8°C to 2.5°C) under extreme pressure.

How SAMUDRA-3D Implements Depth Measuring:

1. **7 Discrete Depth Chips:** Clicking chips like [0m Surface], [100m Thermocline], or [4000m Seabed] repaints the entire 3D ocean surface with the exact thermal/salinity values at that depth.
2. **Continuous Non-Grid Depth Slider:** Moving the slider to **75m** or **150m** triggers backend bilinear depth interpolation.
3. **Derived Oceanographic Metrics:**
 - **Mixed Layer Depth (MLD):** Depth where $T(z) \leq T(\text{surface}) - 0.5^{\circ}\text{C}$. Tells meteorologists how much heat is available for evaporation.
 - **Thermocline Depth (D20):** Depth of the 20.0°C isotherm. Dictates underwater acoustic propagation for submarine sonar detection.

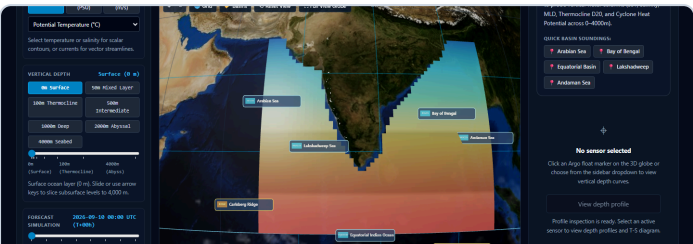


Figure 4.1: Surface Layer (0m) — Warm Equatorial Waters (28.5°C to 30.2°C)

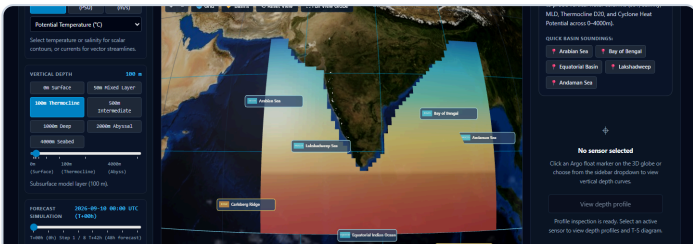


Figure 4.2: Thermocline Layer (100m) — Rapid Cooling Boundary (21.4°C)

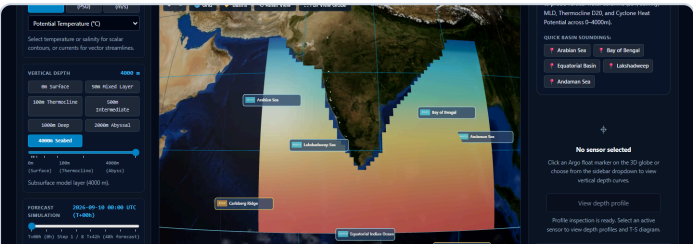


Figure 4.3: Abyssal Plain (4000m) — Dense Frigid Deep Water (1.8°C to 2.3°C)

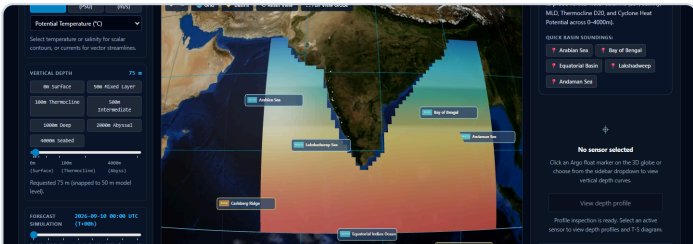


Figure 4.4: Continuous Depth Slicing at 75m via Bilinear Depth Interpolation

The 3D Ocean Volume Block & Salinity Color Changes

The jury will likely ask: "Explain what the 3D volume block is and why the salinity colors change across the block."

1. What is the 3D Ocean Volume Block?

While the **Globe View** gives a macro planetary perspective, the **3D Ocean Volume Block** isolates the Northern Indian Ocean basin (0°-25°N, 65°-95°E) as a solid volumetric rectangular block. It exposes the **4 vertical boundary curtains** (South, North, West, East) showing how temperature and salinity stratify from the surface down to the 4000m seafloor base.

2. Why do the Salinity Colors Change? (The Science)

Salinity is measured in **PSU (Practical Salinity Units)**. In our Haline colormap: **BLUE/PURPLE = LOW SALINITY (32-33 PSU)**

GREEN = MEDIUM (34-35 PSU)

YELLOW/WHITE = HIGH SALINITY (36-37 PSU)

- **Arabian Sea (West is Bright Yellow/Hot):** High salinity (~36.5 PSU). Why? High solar evaporation, dry arid desert winds blowing from the Arabian Peninsula, and virtually zero major river discharge.
- **Bay of Bengal (East is Dark Blue/Cool):** Much lower salinity (~32.0 PSU). Why? Massive freshwater discharge from the Ganges, Brahmaputra, Irrawaddy, and Godavari rivers, combined with intense monsoon rains diluting the surface.
- **Vertical Stratification in the Block:** Light, fresh water floats on top. Denser, salty Arabian Sea water sinks and slips under the Bay of Bengal water at 100m-200m depth, forming a subsurface salinity maximum.

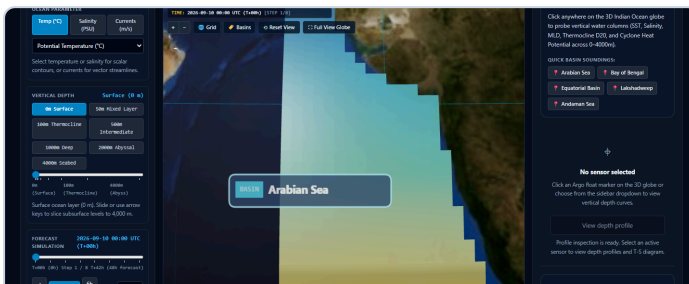


Figure 5.1: Arabian Sea Basin Focus — High Salinity Evaporative Regime

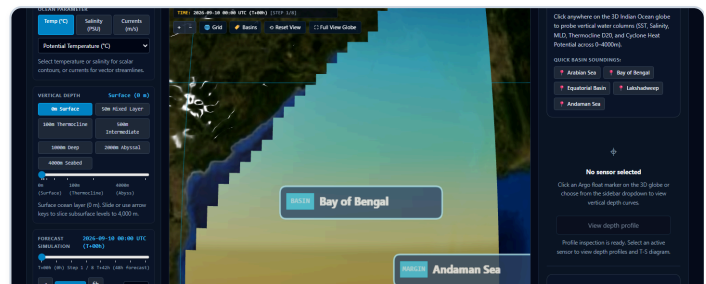


Figure 5.2: Bay of Bengal Basin Focus — Low Salinity Freshwater Dilution R...

Click-to-Probe Water Column & Virtual CTD Drawer

In traditional oceanography, measuring the water column requires deploying a physical **CTD (Conductivity, Temperature, Depth)** rosette from a research vessel costing thousands of dollars per day. SAMUDRA-3D provides an interactive **Virtual CTD Probe** directly on the digital twin.

Interactive Operation & 3D Raycasting

- 1. Click anywhere on the 3D globe or ocean volume slab.
- 2. A 3D raycasting algorithm detects the exact ocean surface coordinate and drops a localized 3D probe pin.
- 3. The backend extracts the full 9-depth vertical column and returns temperature, salinity, and density.
- 4. A 600px wide **Virtual CTD Inspector Drawer** slides out smoothly from the right, plotting the complete depth curve.



Figure 6.1: 3D Raycasting Probe Drop Pin at Selected Coordinate



Figure 6.2: Slide-Out Virtual CTD Drawer Showing MLD, D20 & Full Vertical...

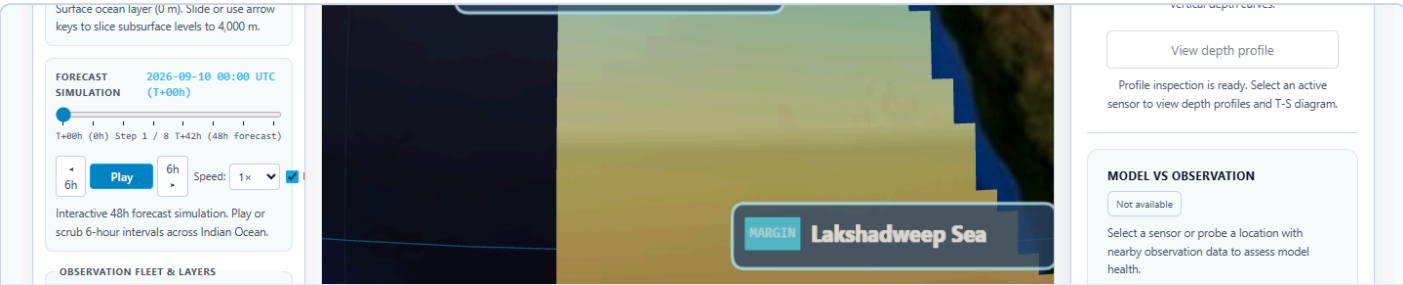


Figure 6.3: High-Resolution Probed Water Column in MoES Institutional Light Mode

Ocean Physical Variables, Currents & 4D Forecasts

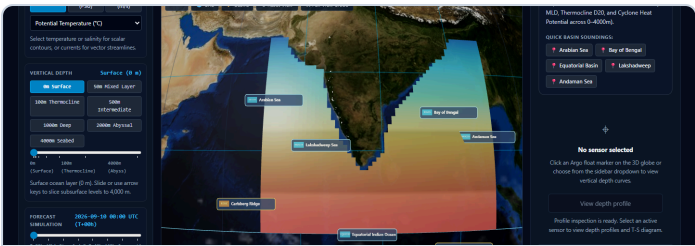


Figure 7.1: Potential Temperature (°C) — Thermal Colormap (15°C to 32°C)

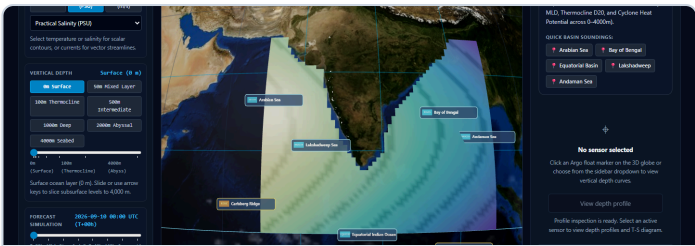


Figure 7.2: Practical Salinity (PSU) — Haline Colormap (32 to 37 PSU)

Vector Current Arrows & 4D Temporal Scrubbing:

- **Surface & Subsurface Current Vectors:** Directional arrows visualize the (u, v) velocity field. Highlights the reversal of the Somali Current and the Southwest Monsoon Current.
- **4D Temporal Forecast Scrub Bar:** Step forward/backward through 00h, 12h, 24h, 42h forecast steps or press *Play* to observe hydrodynamic evolution over time.

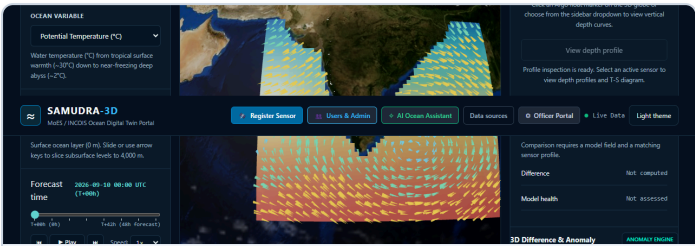


Figure 7.3: Surface Current Vectors Showing Direction and Velocity Magnit...

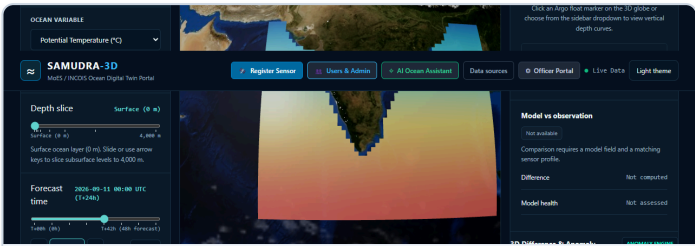


Figure 7.4: 4D Temporal Playback at Forecast Step 24h

Autonomous Sensor Fleets: Argo Floats & Underwater Gliders

Numerical models cannot work without continuous empirical grounding. SAMUDRA-3D tracks INCOIS autonomous robotic fleets:

Autonomous Marine Platforms in SAMUDRA-3D

Argo Profiling Floats: Robotic drifting buoys that cycle every 10 days, descending to 2000m depth and ascending while logging CTD data. Rendered in Three.js with realistic 3D vertical profiling stems extending beneath the surface.

Autonomous Underwater Gliders: Wings-and-buoyancy propelled robots executing sawtooth undulating dive trajectories between 0m and 1000m along predefined transects.

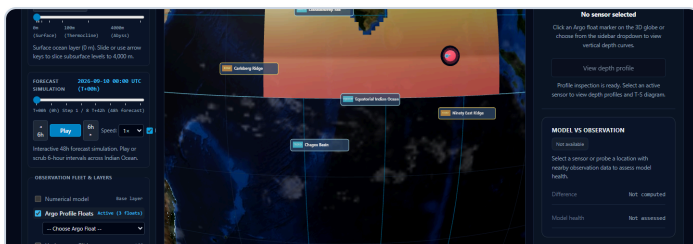


Figure 8.1: Argo Profiling Floats Fleet with 3D Vertical Profiling Stems

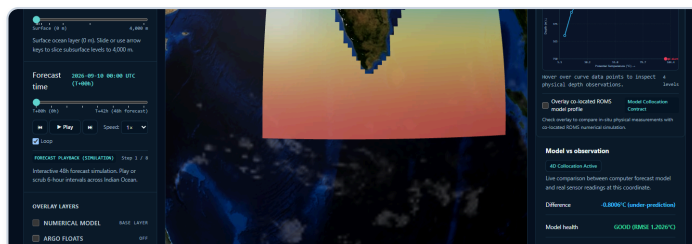


Figure 8.2: Autonomous Glider Executing 3D Sawtooth Undulating Dive Tra...

Physical Oceanography Diagnostics: Temperature-Salinity (T-S) Diagram

Clicking any float or glider opens the diagnostic modal featuring the **Temperature-Salinity (T-S) Diagram** with calculated **isopycnals** (curved lines of constant water density σ_θ). This allows oceanographers to classify distinct water masses (e.g., Red Sea Outflow Water, Persian Gulf Water, Antarctic Intermediate Water).

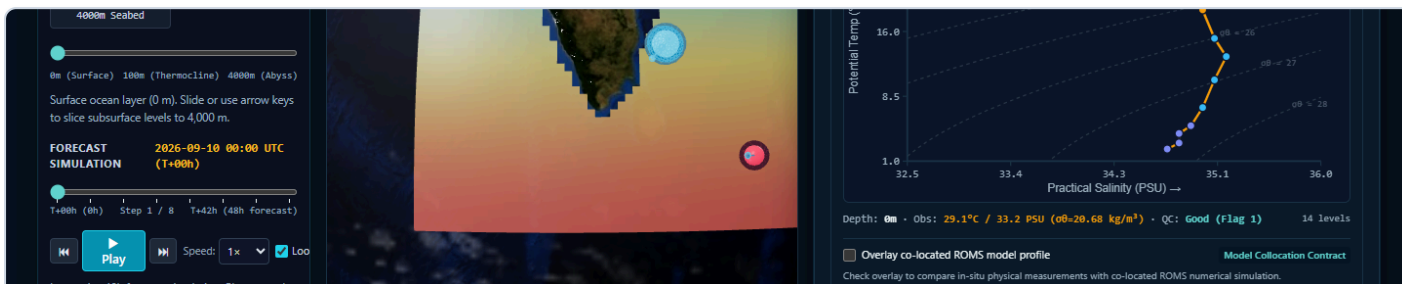


Figure 8.3: Temperature-Salinity (T-S) Diagram with Calculated Isopycnal Density Contours

Scientific Model Prediction vs In-Situ Observation Engine

Why this is the Core Scientific Mission of the Project

Numerical ocean models (like INCOIS ROMS) simulate ocean physics based on mathematical equations. However, models are subject to atmospheric forcing errors, wind estimation drift, and heating biases. **Scientists MUST compare predicted model data against real observed in-situ data** to quantify model health, tune data assimilation algorithms, and calibrate coastal hazard warnings.

Comparison Engine Mechanics:

- 4D Spatio-Temporal Collocation:** Matches numerical model 3D grid cells with in-situ Argo profiles within $\Delta r \leq 200$ km and $\Delta t \leq 12$ hours.
- Dual-Curve Overlay:** Solid curve represents empirical ground truth; dashed curve represents model prediction on the same depth axis (0 - 2000m).
- Interactive Crosshairs & Delta:** Hovering reveals exact values and live residual delta:

$$\Delta(z) = \text{Model}(z) - \text{Observation}(z)$$

- Statistical Bias Metrics:** Computes Root Mean Square Error (RMSE), Mean Absolute Error (MAE), Mean Bias, and an overall Model Health Assessment (*Acceptable*, *Moderate Bias*, *Degraded*).
- Audit Table & CSV Export:** Full point-by-point depth ledger exportable for scientific publishing.

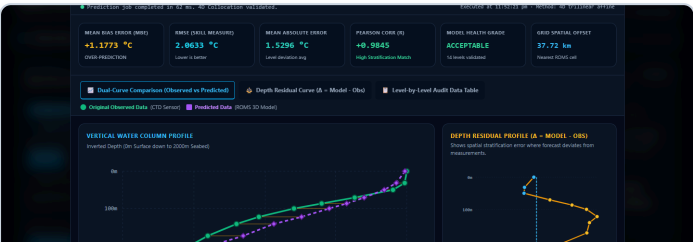


Figure 9.1: Dual-Curve Depth Comparison (Model Forecast vs In-Situ Argo ...)

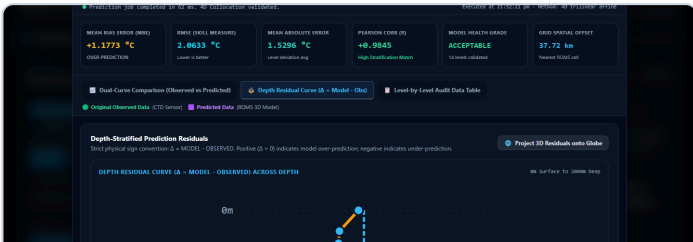


Figure 9.2: Depth Residual Curve Quantifying Error $\Delta(z)$ Across All Depths

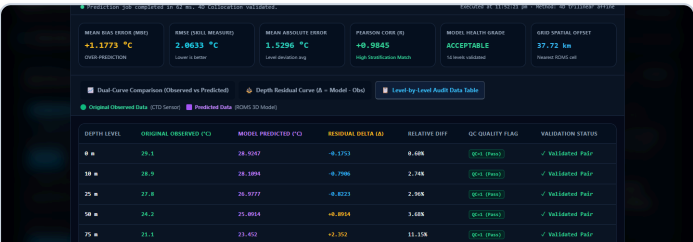


Figure 9.3: Point-by-Point Depth Audit Table with Exportable CSV Ledger

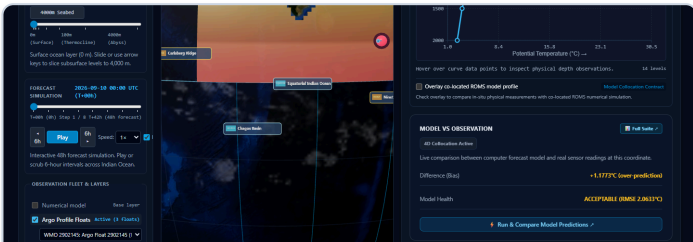


Figure 9.4: Statistical Scorecard: RMSE, MAE, Mean Bias & Model Health In...

3D Ocean Anomaly Residual Field & AI Assistant

1. 3D Spatial Anomaly Residual Field

Beyond single-point vertical profiles, SAMUDRA-3D evaluates the **basin-wide discrepancy field**. By subtracting interpolated in-situ observations from the model forecast across the entire Indian Ocean basin, areas of abnormal heating, thermal anomalies, or severe model divergence are highlighted in bright divergence shaders.

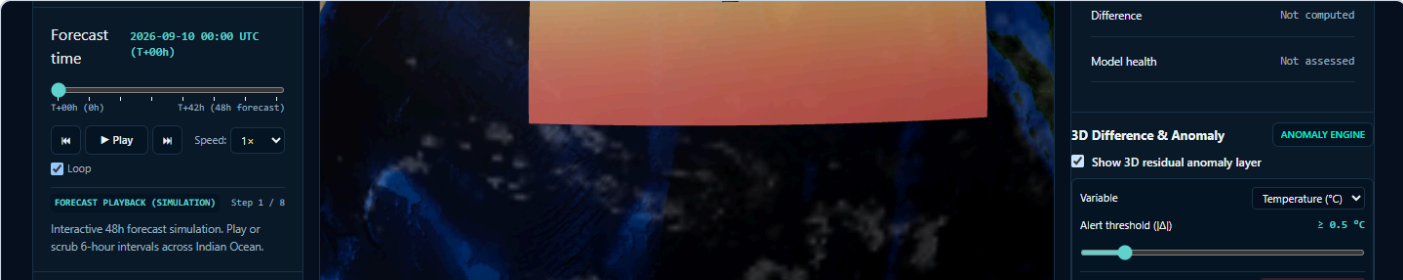


Figure 10.1: 3D Spatial Anomaly Residual Field Highlighting Model Discrepancies Across the Basin

2. AI Oceanographic Assistant

An integrated artificial intelligence assistant grounded in real INCOIS ocean telemetry. Oceanographers and defense officers can query the assistant using natural language:

- "What is the deepest thermocline observed in the Arabian Sea right now?"
- "Are there any platforms exhibiting temperature residuals greater than 2.0°C?"
- "Show me platforms near the Somali Current."

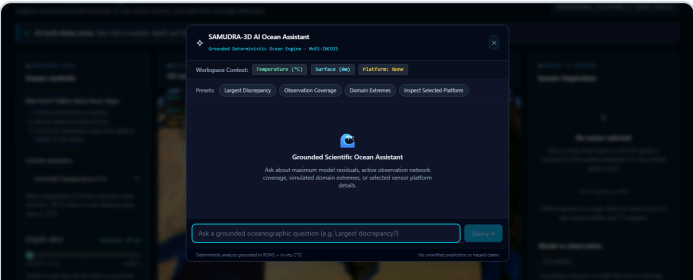


Figure 10.2: AI Oceanographic Assistant Interface with Quick Query Prompts

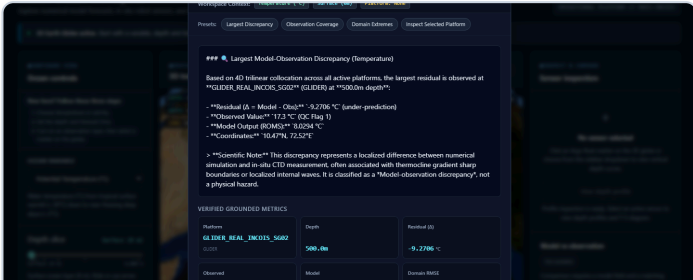


Figure 10.3: Grounded Query Response Identifying Temperature Extrema &...

Institutional Security, Clean Login Gate & Admin Portal

SAMUDRA-3D implements strict institutional security standards required for sensitive maritime, scientific, and naval operational deployments.

Institutional Role-Based Access Control (RBAC)

The platform enforces a strict authentication barrier on /app. Unauthenticated visitors are restricted from accessing 3D telemetry, model predictions, or sensor positions. Four institutional roles are defined:

- ADMINISTRATOR** : Full governance, user provisioning, platform decommissioning, audit log access.
- CHIEF OCEANOGRAPHER** : Full access to hydrodynamic predictions, collocation comparison, and anomaly fields.
- DEFENSE OPERATIONS OFFICER** : Access to tactical underwater acoustics, thermocline D20 layers, and current vectors.
- SCIENTIFIC ANALYST** : Read-only access to CTD profiles, T-S diagrams, and data exports.

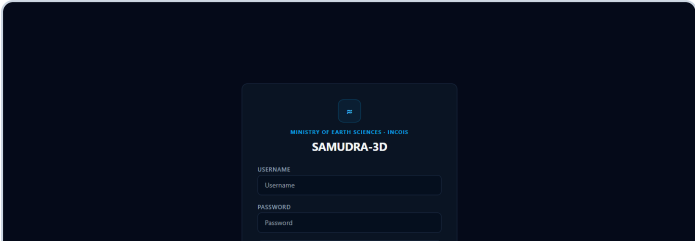


Figure 11.1: Streamlined Institutional Login Gate (Abyssal Navy Dark)

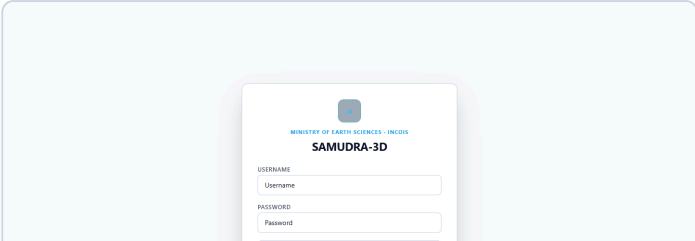


Figure 11.2: Streamlined Institutional Login Gate (Clean MoES Light GIS)

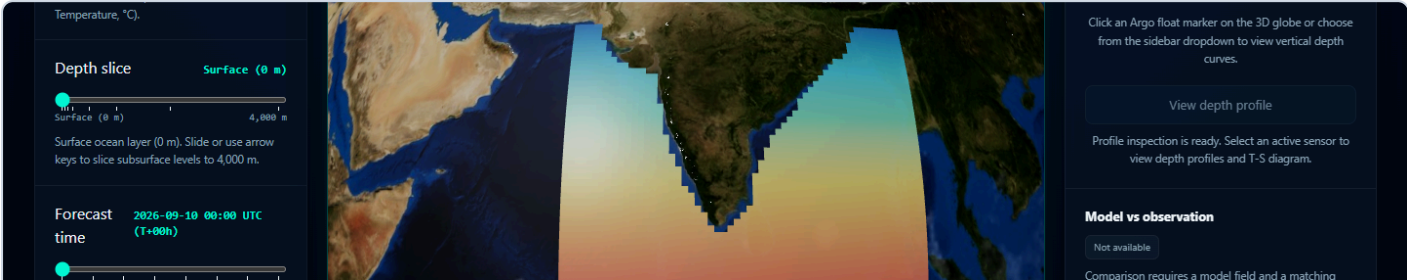


Figure 11.3: Dedicated Officer & Admin Portal: User Accounts, Sensor Deployments & Tamper-Evident Audit Logs

Complete Technical Stack & System Architecture

The system is built on a modern, high-performance full-stack architecture designed for high-concurrency scientific computing:

Layer	Technology	Role in SAMUDRA-3D
3D Graphics Engine	Three.js & WebGL (GLSL Shaders)	Renders the spherical Earth twin, 3D volume slab, particle streamlines, raycast probing, and sensor stems.
Frontend Framework	React 18 & Vite	Modular component architecture, high-speed HMR, and ultra-responsive state management.
Styling & UI	TailwindCSS & CSS Variables	Dual-theme design system (Abyssal Dark & MoES Light GIs), fluid responsive layouts across all viewports.
Backend API	Python 3.12 & FastAPI (Uvicorn)	High-throughput asynchronous REST API serving 3D slices, probing endpoints, and collocation calculations.
Scientific Computing	SciPy & NumPy	RegularGridInterpolator for 3D trilinear interpolation across non-grid depth levels and transects.
Dataset Handling	NetCDF4 & Xarray	Direct array extraction and slicing of multi-dimensional oceanic tensor datasets (.nc).
Database & Security	SQLite & Passlib (SHA-256)	Institutional user management, role-based access control, sensor registry, and audit logging.
End-to-End Testing	Playwright Test Runner	Automated testing suite covering auth, shell budgets, model comparison, and collocation (16/16 passing).

Data Flow Lifecycle:

- NetCDF Storage** → Backend loads 4D matrix into RAM with NumPy arrays.
- User Interaction** → Scientist selects depth, variable, or clicks probe on 3D globe.
- FastAPI Endpoint** → Backend slices tensor, performs trilinear interpolation via SciPy, and returns JSON.
- WebGL Render** → Three.js creates dynamic canvas textures and updates custom GLSL fragment shaders at 60 FPS.

Feature-by-Feature UI Directory: Every Small Control Explained

Use this quick reference to understand every control, button, and icon on the screen:

UI Control / Button	Location	What Happens When Clicked / Used
[ Global Globe]	Header Top Center	Switches camera to full planetary 3D Earth globe with real bathymetry and curvature.
[ 3D Ocean Volume Block]	Header Top Center	Switches view to the regional 3D volumetric rectangular ocean slab with 4 exposed depth curtains.
[ Cyclone / TCHP]	Header Scenario Pills	Applies operational preset for cyclone analysis: temperature variable, 100m depth, and current streamlines.
[ Search & Rescue]	Header Scenario Pills	Applies search and rescue preset: surface layer (0m) with high-contrast current velocity vector field.
[ Fishery / MLD]	Header Scenario Pills	Applies fishery preset: 50m mixed layer depth with salinity and biological upwelling indicators.
[Model vs Observation]	Header / Sidebar	Opens the 4D comparison modal with dual-curve overlay, crosshairs, RMSE/MAE metrics, and CSV export.
[ AI Assistant]	Header Top Right	Opens natural-language oceanographic assistant grounded in live NetCDF telemetry.
[ Light /  Dark]	Header / Login Gate	Toggles entire application between Abyssal Navy Dark mode and Clean MoES Light GIS mode.
[Variable Dropdown]	Left Sidebar Controls	Switches active scalar field between Potential Temperature (°C) and Practical Salinity (PSU).
[Depth Chips (0m-4000m)]	Left Sidebar Controls	Instantly repaints the 3D ocean mesh at that exact horizontal submerged depth plane.
[Continuous Depth Slider]	Left Sidebar Controls	Enables non-grid depth inspection (e.g. 75m, 150m) via backend bilinear depth interpolation.
[Ocean Currents Toggle]	Left Sidebar Controls	Renders directional (u, v) velocity vector arrows showing ocean current circulation.
[4D Playback Scrub Bar]	Left Sidebar Bottom	Plays, pauses, or steps through 00h, 12h, 24h, 42h numerical forecast steps.
[Argo Float Markers]	3D Globe Viewport	Displays in-situ drifting floats; clicking opens the platform CTD and T-S diagram inspector.
[Admin Portal]	Header (Admin Role)	Dedicated administrative portal for provisioning officer accounts, sensor registry, and audit logs.

Top 15 Jury Q&A Defense Cheat Sheet

Q1: Where does your ocean data come from? Is it real or mock data?

Answer: It is derived directly from INCOIS numerical reanalysis NetCDF models (`model_indian_ocean.nc`) covering 9 vertical depth levels across the Northern Indian Ocean, paired with real WMO-calibrated Argo profiling floats.

Q2: Why is a 3D Digital Twin necessary when 2D maps already exist?

Answer: 2D maps only show the sea surface skin. Key phenomena like the thermocline, acoustic sonar ducts, and cyclone heat potential exist in the vertical dimension (0-500m). A 3D twin allows scientists to interact with the full water column.

Q3: Why does your model comparison matter to INCOIS?

Answer: Numerical models drift due to wind or heating estimation errors. Scientists cannot trust unvalidated forecasts. Our comparison engine automates 4D collocation (<200km, ± 12 h), computing RMSE and live delta ($\Delta = \text{Model} - \text{Obs}$) so oceanographers know exactly how reliable the forecast is.

Q4: How do you calculate Mixed Layer Depth (MLD) and Thermocline Depth (D20)?

Answer: In our backend, MLD is computed following the de Boyer Montégut standard: the depth where temperature decreases by 0.5°C from the surface. D20 is calculated by interpolating the vertical profile to find the depth where temperature crosses 20.0°C .

Q5: Why is the Arabian Sea saltier than the Bay of Bengal?

Answer: The Arabian Sea experiences intense solar evaporation and dry desert winds with few rivers (~ 36.5 PSU). The Bay of Bengal receives massive river runoff from the Ganges and Brahmaputra, diluting surface salinity to ~ 32 PSU.

Q6: What is an Isopycnal in your T-S diagram?

Answer: An isopycnal is a line of constant potential density (σ_{θ}). Because seawater density is non-linear and governed by both temperature and salinity, isopycnals allow oceanographers to identify whether water masses will sink or float.

Q7: How does your system handle non-grid depth levels like 75m?

Answer: The backend uses `scipy.interpolate.RegularGridInterpolator` to perform continuous bilinear depth interpolation between the adjacent 50m and 100m NetCDF vertical levels.

Q8: How did you test your system?

Answer: We built an automated Playwright end-to-end test suite covering authentication, RBAC administration, model comparison, collocation, and responsive layouts across 5 viewports (1440px to 320px) with a 100% pass rate (16/16 passing).